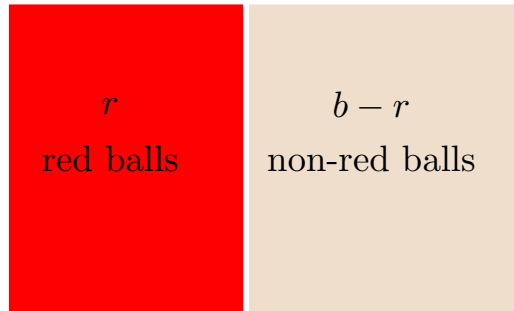


Homework 2- Due Date Feb 26, 11:59 P.M.

Problem 1. A box contains b balls, out of which exactly r are red. We select k of the balls at random, without replacement. What is the probability that m of the selected balls are red?

Solution.



$|S|$ = choose k balls from b balls = $\binom{b}{k}$

E = choose k balls such that m are red and $k - m$ are nonred = That is choose m balls from r red balls and choose $k - m$ balls from $b - r$ balls. Hence,

$$P(E) = \frac{\binom{r}{m} \binom{b-r}{k-m}}{\binom{b}{k}}$$

Problem 2. Out of eight women and eight men, we form a committee consisting of six different people. Assuming that each committee of size six is equally likely, find the probabilities of the following events:

- (a) The committee consists of three men and three women.
- (b) The committee has at least one woman.
- (c) For the remainder of the problem, assume that Emma and Jack are among the sixteen people being considered. Both Emma and Jack are members of the committee.

Solution. $|S|$ = choose 6 people from 16 people.

(a) Event A = choose 3 women from 8 women and choose 3 men from 8 men

$$P(A) = \frac{\binom{8}{3} \binom{8}{3}}{\binom{16}{6}}$$

(b) Event B = no woman is chosen, that is all 6 people are chosen from men.

$$P(B^c) = 1 - P(B) = 1 - \frac{\binom{8}{6}}{\binom{16}{6}}$$

(c) Event C= we want to form a group of 6, which two of them are Emma and Jack, so we need to choose the rest of 4 people from the rest of 14 people.

$$P(C) = \frac{\binom{14}{4}}{\binom{16}{6}}$$

Problem 3. We want to arrange 3 novels, 2 mathematics books, and 1 chemistry book on a bookshelf

- (b) What is the probability that the mathematics books are together and the novels are together?
- (c) What is the probability that the novels are together, but the other books can be arranged in any order?

Sample space = S = {All the possible orderings of the book, with no restriction}. So in total there are 6 books, so $|S| = 6!$.

(b) A = {Order the book in the way books with same subjects are together}. In total we have 3 subjects, so $3!$ ordering for subjects. But math books can be arranged in $2!$ way, novel books in $3!$, and chemistry book in $1!$. so in total

$$|A| = \underbrace{3!}_{\text{subjects}} \cdot \underbrace{3!}_{\text{novel}} \cdot \underbrace{2!}_{\text{math}} \cdot \underbrace{1!}_{\text{chemistry}}$$

Therefore,

$$P(A) = \frac{3! \cdot 3! \cdot 2! \cdot 1!}{6!}$$

(c) B= Only novel books are together and rest of books can be arranged in any order. Similar to above assume all novels are one pile of books, so in total we have one pile and 3 more books. so in total 4 things that need to be rearranged hence $4!$, but within our pile of novel we can arrange books in $3!$ ways hence in total we have $4!3!$ way of rearranging.

$$P(B) = \frac{4! \cdot 3!}{6!}$$

Problem 4. Suppose $P(A) = 0.4$, $P(B') = 0.3$, and $P(A \cap B') = 0.1$.

(a) Find $P(A \cup B)$.

(b) Find $P(B'|A)$.

(c) Find $P(B|A')$.

Solution.

$$(a) P(A \cap B) = P(A) - P(A \cap B') = 0.4 - 0.1 = 0.3$$

$$P(B) = 1 - P(B') = 1 - 0.3 = 0.7$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.7 - 0.3 = 0.8$$

$$(b) P(B'|A) = \frac{P(B' \cap A)}{P(A)} = \frac{0.1}{0.4} = \frac{1}{4}$$

(c)

$$P(A \cap B) = P(A) - P(A \cap B') = 0.4 - 0.1 = 0.3$$

$$P(A' \cap B) = P(B) - P(A \cap B) = 0.7 - 0.3 = 0.4$$

$$P(A') = 1 - P(A) = 1 - 0.4 = 0.6$$

$$P(B|A') = \frac{P(A' \cap B)}{P(A')} = \frac{0.4}{0.6} = \frac{2}{3}$$

Problem 5. We have an infinite collection of biased coins, indexed by the positive integers. Coin n has probability $\frac{2}{3^n}$ of being selected. A flip of coin n results in Tail with probability $\frac{1}{2^n}$. We select a coin and flip it. What is the probability that the result is Tail?

The probability that the result is Tail is:

Solution.

$$P(\text{coin } n) = \frac{2}{3^n} \text{ and } P(\text{Tail}|\text{coin } n) = \frac{1}{2^n}$$

Using law of total probability (LOTP)

$$P(\text{Tail}) = \sum_{n=1}^{\infty} P(\text{Tail}|\text{coin } n)P(\text{coin } n) = 2 \sum_{n=1}^{\infty} \left(\frac{1}{6}\right)^n$$

$\left(\frac{1}{6}\right) < 1$, by using properties of geometric series

$$P(\text{Tail}) = 2 \cdot \frac{\frac{1}{6}}{1 - \frac{1}{6}} = 2 \cdot \frac{\frac{1}{6}}{\frac{5}{6}} = \frac{2}{5}$$

Problem 6. It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%.

Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?

Define events

A = event that an email is detected as spam,

B = event that an email is spam,

B' = event that an email is not spam.

We know $P(B) = P(B') = 0.5$, $P(A|B) = 0.99$, $P(A|B') = 0.05$. Hence by the Bayes's formula we have

$$\begin{aligned} P(B'|A) &= \frac{P(A|B')P(B')}{P(A|B)P(B) + P(A|B')P(B')} \\ &= \frac{0.05 \times 0.5}{0.05 \times 0.5 + 0.99 \times 0.5} = \frac{5}{104} \end{aligned}$$

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